

SQL Command Types

DDL	DQL	DML	DCL	TCL
CREATE DROP ALTER TRUNCATE COMMENT RENAME	SELECT	INSERT UPDATE DELETE LOCK MERGE	GRANT REVOKE	COMMIT ROLLBACK SAVEPOINT SET TRANSACTION

What is SQL ?

- SQL stands for Structured Query Language.
- SQL is a standard language for relational database management systems.
- SQL is specially useful in handling organized data comprised of entities and relations between different entities of data.





What is DBMS ?

- Database management system is a software that is used to manage the database.
- A DBMS serves as an interface between end-user and database, allowing users to

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and a database, allowing user to create, read, delete data in the database.
. Most widely used types of DBMS software are relation, distributed, hierarchical, object-oriented and network.

3] What is Database?

. A database is an organized collection of data, and it can be easily accessed & managed.
. We can organize data into tables, rows columns and index to find relevant info.
. The main purpose of the database is to operate large amount of information by storing, retrieving and managing data.
. There are many databases available like MySQL, Oracle, MongoDB, SQL server.

4] What is RDBMS?

→ . RDBMS stands for Relational Database Management system
. Relational Database Management system is collection of programs & compatibilities that enables IT teams.
. RDBMS store data in the form of tables.

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.. RDBMS provides a dependable method of storing and retrieving large amount of data.
. RDBMS uses SQL queries to access the data in database.

5] What is a Database Table?

→ . A table is a collection of related data entries and consist of column & rows.
. A row is each individual entry that exist in a table.
. A column holds specific information about every record in a table.
. Columns can be arranged in verticle and rows as horizontal.
. The column in table also called fields and rows called records.

6] What is Primary key?

→ . A primary key is a minimal set of at attributes in a table that uniquely identify rows in that table i.e. tuples.
. When multiple columns are used as a primary key, it is known as composite primary key.
. While using composite primary key you should use few column as possible.

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- When more columns you use for primary key the more storage space you required.
- Primary key has unique data.
- Primary key cannot have null value
- We cannot use duplicate value for primary key.
- Maximum length of primary key is 900 bytes.

7] What is Foreign key?

→ - A foreign key is a key used to link two tables together. it also called referencing key.
- A foreign key is a combination of column whose value match primary key in different table.
- Use of foreign key is to link attributes of two tables with the help of primary key. attribute.
- Hence, it used for creating & maintaining the relationship bet^n two relations.

8] What is a Join?

→ - The SQL join clause is used to combine rows from two or more tables in a SQL database based on related column between the two.

. There are four different types of JOIN's:

1) JOIN
2) LEFT JOIN
3) RIGHT JOIN
4) FULL JOIN.

9] What is the SELECT statement?

→ . SELECT statement in SQL is used to recover data from a database.
. The data returned is stored in result table called result-set.
. The SELECT to determine what to include in Final result set.

10] What is cursor in SQL?

→ . Cursor is a temporary work station.
. It created in system memory When SQL statement is executed.
. A SQL cursor is a set of rows together with points that identify rows.
. Types of cursor:
  1) Implicit cursor.
  2) Explicit cursor.
. Cursor are used to store Database tables.

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11] What are Entities?

→ . An entity can be real world object.
. An entity can be a single thing, person, place or object.
. Data can be stored about such entities.

12] What is relationships in SQL?

→ . Relationship is a link between entities that have something to do with each other.
. It includes one to one, many to many, many to one relations.
. Relation in database allow for clear table & help cut redundant database is important.

13] What is an index is SQL?

→ . SQL index is a table used to speed up the searching of the data.
. It also recover large amount of data from tables.
. INDEX requires its own space in storage device.
. INDEX is a pointer to data in a table.
. An INDEX is same to an index in the back of book.
. INDEX is also unique constraint.

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14] What is a Self-Join?

→ · It is a join in which table is joined with itself.
· The self join can be viewed as a join of two copies of same table.
· Self-join statement is important when two sets of data in same table are compared.

15] What is a cross-Join?

→ · Cross-Join is a Cartesian product of rows from tables in join.
· Cross Join is combination of each row of first ~~column~~ table with each row of second table.
· It is also called cartesian join.

16] What is Aliase in SQL?

→ · Aliases are temporary name given to table or column for SQL ~~se~~ query.
· It used when name of table is used other than original name.
· Aliases used to make column names more readable.
· Aliases useful when there are more than one table complicated in a query.

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17] **What is Normalization?**

→ · Normalization is a process restructure the data in database.
· There is no redundancy of data, all data stored in one place.
· It eliminates characteristics like Insertion, Update and Deletion Anomalis.
· Normalization converts large tables in small tables. and links them by relationships.

18] **What is Denormalization?**

→ · It is inverse process of Normalization.
· When converted all small tables want to make single table then we have to use denormalization.
· Denormalization is a technique used by database to optimize the efficiency of database.
· This method allows to add mixed data into normal database.



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